A Level • Cambridge (CIE) • Chemistry

41 mins 22 questions

Multiple Choice Practice

Topical Practice MCQ

5. Chemical energetics

Carefully selected past paper questions by topic.

9 The enthalpy change for a reaction can be calculated from values of:

- enthalpies of formation, $\Delta H_f^{\ominus}$
- enthalpies of combustion, $\Delta H_c^{\ominus}$
- bond energies, E.

The enthalpy change of the reaction given = $\Delta H_r^{\ominus}$.

$$2C_2H_6(g) + 3O_2(g) \rightarrow 2CH_4(g) + 2CO_2(g) + 2H_2O(l)$$

Which expression could be used to calculate $\Delta H_r^{\ominus}$?

A $\Delta H_c^{\ominus}(C_2H_6(g))$

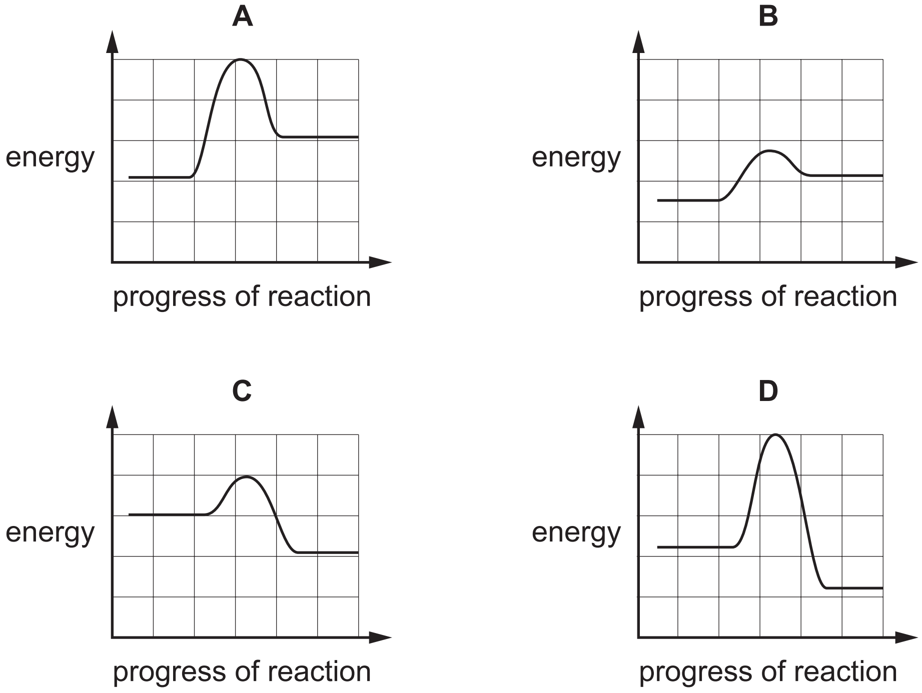
B $2\Delta H_c^{\ominus}(C_2H_6(g)) - 2\Delta H_c^{\ominus}(CH_4(g))$

C $E(C{-}C) + 2E(C{-}H) - 4E(C{=}O) - 4E(H{-}O)$

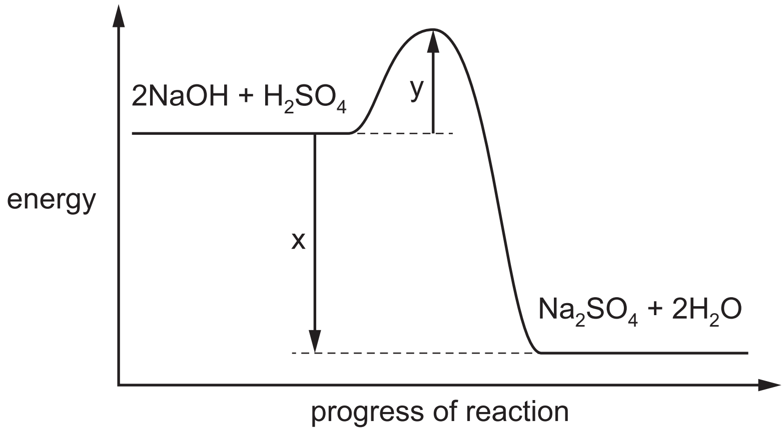
D $\Delta H_f^{\ominus}(CH_4(g)) + \Delta H_f^{\ominus}(CO_2(g)) + \Delta H_f^{\ominus}(H_2O(l)) - \Delta H_f^{\ominus}(C_2H_6(g))$

10 For a certain endothermic reaction, the activation energy is numerically equal to twice the enthalpy change of reaction.

Which reaction pathway diagram is correct for this reaction?



10 A reaction pathway diagram for the reaction of aqueous sodium hydroxide and dilute sulfuric acid is shown.



What is the value of the enthalpy change of neutralisation, ΔH_{neut}?

A x

B x – y

C $\dfrac{x}{2}$

D $\dfrac{(x - y)}{2}$

9 Which equation represents an enthalpy change that is the average bond energy of the C–H bond in methane?

A $\frac{1}{4}C(g) + H(g) \rightarrow \frac{1}{4}CH_4(g)$

B $\frac{1}{4}CH_4(g) \rightarrow \frac{1}{4}C(g) + H(g)$

C $CH_4(g) \rightarrow C(g) + 4H(g)$

D $CH_4(g) \rightarrow CH_3(g) + H(g)$

9 The enthalpy changes of formation, $\Delta H_f^\ominus$, of both PCl_3 and PCl_5 are exothermic.

PCl_3 reacts with chlorine.

$$PCl_3(l) + Cl_2(g) \rightarrow PCl_5(s) \qquad \Delta H^\ominus_{reaction} = -124\,kJ\,mol^{-1}$$

Which pair of statements is correct?

	statement 1	statement 2
A	$\Delta H^\ominus_{reaction}$ is less negative than $\Delta H_f^\ominus$ (PCl_5).	The Cl_2 bond energy is needed in calculating $\Delta H^\ominus_{reaction}$ from enthalpies of formation.
B	$\Delta H^\ominus_{reaction}$ is more negative than $\Delta H_f^\ominus$ (PCl_5).	The Cl_2 bond energy is needed in calculating $\Delta H^\ominus_{reaction}$ from enthalpies of formation.
C	$\Delta H^\ominus_{reaction}$ is less negative than $\Delta H_f^\ominus$ (PCl_5).	The Cl_2 bond energy is not needed in calculating $\Delta H^\ominus_{reaction}$ from enthalpies of formation.
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16 Use relevant enthalpy changes from the tables to answer this question.

reaction	ΔH / kJ mol^{-1}
$C(s) + 2H_2(g) \rightarrow CH_4(g)$	–76
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$	–890
$CH_4(g) \rightarrow C(g) + 4H(g)$	1648
$3C(s) + 4H_2(g) \rightarrow C_3H_8(g)$	–105

bond	bond enthalpy / kJ mol^{-1}
H–H	436
C–C	350
C=C	610
C=O	805

Which value can be calculated for the enthalpy change for the following reaction?

$$2C(g) + 6H(g) \rightarrow C_2H_6(g)$$

A –2822 kJ mol^{-1}

B –2122 kJ mol^{-1}

C –1998 kJ mol^{-1}

D –1772 kJ mol^{-1}

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7 Using the information in the table, what is the enthalpy change, q, for the reaction described?

$$Cs^+(g) + Br^-(g) \rightarrow CsBr(s) \qquad \Delta H = q\,kJ\,mol^{-1}$$

standard enthalpy change	value / $kJ\,mol^{-1}$
$\Delta H^{\ominus}_{sol}$ CsBr(s)	+25.9
$\Delta H^{\ominus}_{hyd}$ $Cs^+(g)$	–276
$\Delta H^{\ominus}_{hyd}$ $Br^-(g)$	–335

A –636.9 **B** –585.1 **C** +585.1 **D** +636.9

8 Which equation represents the enthalpy change of atomisation of iodine?

A $\frac{1}{2}I_2(g) \rightarrow I(g)$

B $I_2(g) \rightarrow 2I(g)$

C $\frac{1}{2}I_2(s) \rightarrow I(g)$

D $I_2(s) \rightarrow 2I(g)$

1 Enthalpy changes, ΔH, can be positive or negative.

Which row is correct?

	ΔH positive	ΔH negative
A	atomisation	bond breaking
B	bond breaking	neutralisation
C	bond making	combustion
D	combustion	bond making

6 Solid sulfur consists of molecules made up of eight atoms covalently bonded together.

The bonding in sulfur dioxide is O=S=O.

enthalpy change of combustion of S_8, $\Delta H^{\ominus}_{c} S_8(s) = -2376\,kJ\,mol^{-1}$

energy required to break 1 mole $S_8(s)$ into gaseous atoms = $2232\,kJ\,mol^{-1}$

O=O bond enthalpy = $496\,kJ\,mol^{-1}$

Using these data, what is the value of the S=O bond enthalpy?

A $239\,kJ\,mol^{-1}$ **B** $257\,kJ\,mol^{-1}$ **C** $319\,kJ\,mol^{-1}$ **D** $536\,kJ\,mol^{-1}$

7 *Use of the Data Booklet is relevant for this question.*

In an experiment, the burning of 1.45 g (0.025 mol) of propanone was used to heat 100 g of water. The initial temperature of the water was 20.0 °C and the final temperature of the water was 78.0 °C.

Which experimental value for the enthalpy change of combustion for propanone can be calculated from these results?

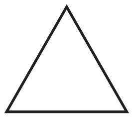
A $-1304\ \text{kJ mol}^{-1}$

B $-970\ \text{kJ mol}^{-1}$

C $-352\ \text{kJ mol}^{-1}$

D $-24.2\ \text{kJ mol}^{-1}$

11 The diagram shows the skeletal formula of cyclopropane.



The enthalpy change of formation of cyclopropane is +53.3 kJ mol^{-1} and the enthalpy change of atomisation of graphite is +717 kJ mol^{-1}.

The bond enthalpy of H–H is 436 kJ mol^{-1} and of C–H is 410 kJ mol^{-1}.

What value for the average bond enthalpy of the C–C bond in cyclopropane can be calculated from this data?

A 187 kJ mol^{-1} **B** 315 kJ mol^{-1} **C** 351 kJ mol^{-1} **D** 946 kJ mol^{-1}

10 *Use of the Data Booklet is relevant to this question.*

A student mixed 25 cm^3 of 0.10 mol dm^{-3} sodium hydroxide solution with 25 cm^3 of 0.10 mol dm^{-3} hydrochloric acid and noted a temperature rise of 2.5 °C.

What is the enthalpy change of the reaction per mole of NaOH?

A –209 kJ mol^{-1}

B –104.5 kJ mol^{-1}

C –209 J mol^{-1}

D –522.5 J mol^{-1}

10 A student calculated the standard enthalpy change of formation of ethane, C_2H_6, using a method based on standard enthalpy changes of combustion.

He used correct values for the standard enthalpy change of combustion of ethane ($-1560\ kJ\ mol^{-1}$) and hydrogen ($-286\ kJ\ mol^{-1}$) but he used an incorrect value for the standard enthalpy change of combustion of carbon. He then performed his calculation correctly. His final answer was $-158\ kJ\ mol^{-1}$.

What did he use for the standard enthalpy change of combustion of carbon?

A $-1432\ kJ\ mol^{-1}$

B $-860\ kJ\ mol^{-1}$

C $-430\ kJ\ mol^{-1}$

D $-272\ kJ\ mol^{-1}$

11 Which process could be used to calculate the bond energy for the covalent bond X-Y by dividing its ΔH by n?

A $XY_n(g) \rightarrow X(g) + nY(g)$

B $2XY_n(g) \rightarrow 2XY_{n-1}(g) + Y_2(g)$

C $Y(g) + XY_{n-1}(g) \rightarrow XY_n(g)$

D $nXY(g) \rightarrow nX(g) + \frac{n}{2}Y_2(g)$

8 The equation below represents the combination of gaseous atoms of non-metal X and of hydrogen to form gaseous X_2H_6 molecules.

$$2X(g) + 6H(g) \rightarrow X_2H_6(g) \qquad \Delta H = -2775\,kJ\,mol^{-1}$$

The bond energy of an X–H bond is $395\,kJ\,mol^{-1}$.

What is the bond energy of an X–X bond?

A $-405.0\,kJ\,mol^{-1}$

B $-202.5\,kJ\,mol^{-1}$

C $+202.5\,kJ\,mol^{-1}$

D $+405.0\,kJ\,mol^{-1}$

10 Hess's Law can be used to calculate the average C-H bond energy in methane.

$\Delta H^\ominus_{at}$ = standard enthalpy change of atomisation

$\Delta H^\ominus_{f}$ = standard enthalpy change of formation

$\Delta H^\ominus_{c}$ = standard enthalpy change of combustion

Which data values are needed in order to perform the calculation?

A $\Delta H^\ominus_{at}$ (C), $\Delta H^\ominus_{at}$ (H), $\Delta H^\ominus_{f}$ (CH_4)

B $\Delta H^\ominus_{c}$ (C), $\Delta H^\ominus_{c}$ (H_2), $\Delta H^\ominus_{c}$ (CH_4)

C $\Delta H^\ominus_{c}$ (C), $\Delta H^\ominus_{c}$ (H_2), $\Delta H^\ominus_{f}$ (CH_4)

D $\Delta H^\ominus_{f}$ (CH_4) only, as $\Delta H^\ominus_{f}$ (C), and $\Delta H^\ominus_{f}$ (H_2), are defined as zero

5 Given the following enthalpy changes,

$$I_2(g) + 3Cl_2(g) \rightarrow 2ICl_3(s) \qquad \Delta H^\ominus = -214\,kJ\,mol^{-1}$$

$$I_2(s) \rightarrow I_2(g) \qquad \Delta H^\ominus = +38\,kJ\,mol^{-1}$$

What is the standard enthalpy change of formation of iodine trichloride, $ICl_3(s)$?

A $+176\,kJ\,mol^{-1}$

B $-88\,kJ\,mol^{-1}$

C $-176\,kJ\,mol^{-1}$

D $-214\,kJ\,mol^{-1}$

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B $-88\,kJ\,mol^{-1}$

C $-176\,kJ\,mol^{-1}$

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11 Which equation represents the change corresponding to the enthalpy change of atomisation of iodine?

A $\frac{1}{2}I_2(g) \rightarrow I(g)$

B $I_2(g) \rightarrow 2I(g)$

C $\frac{1}{2}I_2(s) \rightarrow I(g)$

D $I_2(s) \rightarrow 2I(g)$

8 *Use of the Data Booklet is relevant to this question.*

The enthalpy change of formation, ΔH_f, of hydrated calcium ions is the enthalpy change of the following reaction.

$$Ca(s) + aq - 2e^- \rightarrow Ca^{2+}(aq)$$

The following enthalpy changes are **not** quoted in the *Data Booklet*.

$$Ca(s) \rightarrow Ca(g) \qquad \Delta H_a = 177\,kJ\,mol^{-1}$$

$$Ca^{2+}(g) + aq \rightarrow Ca^{2+}(aq) \qquad \Delta H_{hyd} = -1565\,kJ\,mol^{-1}$$

What is the enthalpy change of formation of hydrated calcium ions?

A $-1388\,kJ\,mol^{-1}$

B $-798\,kJ\,mol^{-1}$

C $-238\,kJ\,mol^{-1}$

D $+352\,kJ\,mol^{-1}$

Answer Key

No.	Paper	Topic	Answer
1	9701_m24_qp_12	5. Chemical energetics	**B**
2	9701_s24_qp_11	5. Chemical energetics	**B**
3	9701_s22_qp_11	5. Chemical energetics	**C**
4	9701_s22_qp_13	5. Chemical energetics	**B**
5	9701_w22_qp_11	5. Chemical energetics	**C**
6	9701_w22_qp_12	5. Chemical energetics	**A**
7	9701_w22_qp_13	5. Chemical energetics	**C**
8	9701_s21_qp_13	5. Chemical energetics	**A**
9	9701_w21_qp_11	5. Chemical energetics	**C**
10	9701_s16_qp_13	5. Chemical energetics	**B**
11	9701_w15_qp_11	5. Chemical energetics	**D**
12	9701_w15_qp_11	5. Chemical energetics	**B**
13	9701_s14_qp_12	5. Chemical energetics	**B**
14	9701_s13_qp_12	5. Chemical energetics	**A**
15	9701_w12_qp_11	5. Chemical energetics	**C**
16	9701_w12_qp_12	5. Chemical energetics	**A**
17	9701_s11_qp_11	5. Chemical energetics	**D**
18	9701_w11_qp_12	5. Chemical energetics	**A**
19	9701_s10_qp_11	5. Chemical energetics	**B**
20	9701_s10_qp_12	5. Chemical energetics	**B**
21	9701_w10_qp_11	5. Chemical energetics	**C**
22	9701_w10_qp_12	5. Chemical energetics	**D**